

Notes: Hayek (AER, 1945)

The Use of Knowledge in Society

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1 Big picture

- **Question.** What is the real economic problem society must solve?
- **Core answer.** The economic problem is not merely allocating “given” resources. It is using knowledge that is dispersed, local, temporary, incomplete, and often not formally articulable.
- **Main target.** Hayek criticizes the idea that society’s economic problem can be solved as if all relevant data were available to one mind.
- **Key distinction.** Scientific or technical knowledge is only one type of knowledge. Much economically valuable knowledge is knowledge of “particular circumstances of time and place.”
- **Planning reinterpretation.** All economic activity is planning. The real issue is whether planning is centralized in one authority or decentralized among many individuals.
- **Central mechanism.** The price system communicates compressed information. Prices let people adjust to changes whose causes they do not know and do not need to know.
- **Why decentralization matters.** The people closest to local circumstances can react fastest, but they still need signals about broader scarcity. Prices solve this coordination problem.
- **Why equilibrium analysis can mislead.** Equilibrium conditions assume away the problem when they treat all facts as already known to a single mind.
- **Why markets are “marvelous.”** They coordinate many actions without central orders and without requiring each participant to understand the whole system.
- **Economic takeaway.** The price system is an information and coordination technology. Its value is not perfection, but economy of knowledge.

2 Conceptual objects and measurement

2.1 The wrong problem: allocation from given data

In the standard optimum problem, one imagines:

complete information + given preferences + known means $\Rightarrow$ logical allocation problem.

The solution can be described by equilibrium-like conditions:

$$MRS_{ij}^a = MRS_{ij}^b \quad \text{across all uses and agents.}$$

Interpretation: This is a useful logical benchmark, but it is not the social economic problem. It assumes that the data have already been collected, integrated, and understood.

2.2 The real problem: dispersed knowledge

Let individual i possess local knowledge

$$\mathcal{K}_i = \{\text{local scarcities, skills, opportunities, preferences, timing, special circumstances}\}.$$

The social knowledge set is

$$\mathcal{K} = \bigcup_i \mathcal{K}_i,$$

but $\mathcal{K}$ is not possessed by any single mind.

Interpretation: The economic problem is to make use of $\mathcal{K}$ without first converting it into a centralized database. This is why the mechanism of communication matters.

2.3 Kinds of knowledge

Hayek distinguishes:

- **scientific knowledge:** general principles, techniques, and expert knowledge;
- **local knowledge:** circumstances of time and place;
- **practical knowledge:** experience, judgment, and tacit knowledge from occupation and context;
- **temporary opportunity knowledge:** underused machines, available skills, excess inventories, transport opportunities, and local price discrepancies.

Interpretation: Central experts may be good at collecting scientific knowledge. They are much worse at collecting every small fact that matters for day-to-day adaptation.

2.4 Planning

Hayek uses planning broadly:

$\mathcal{P}$ = interrelated decisions about resource allocation.

The question is not whether planning exists. The question is:

central planning by one authority versus decentralized planning by many persons.

Interpretation: Markets are not “unplanned.” They are a system of decentralized plans coordinated by prices.

2.5 Change

Hayek stresses that economic problems arise because circumstances change:

economic problem $\iff$ need to adapt to change.

If nothing changed, or if everything changed exactly as expected, no new economic decision would be needed.

Interpretation: Statistical aggregates can appear stable while countless local adjustments are taking place underneath. The important knowledge is often exactly what aggregate statistics abstract away from.

3 Mechanism: prices as communication

3.1 The man on the spot

The person closest to a decision often knows:

- which local input is scarce;
- which machine is idle;
- which worker has a specialized skill;
- which stockpile can be drawn down;
- which delivery route or temporary opportunity is available.

But he does not know the whole economy. He needs only enough information to fit his action into the broader pattern.

Interpretation: Decentralization uses local knowledge; prices connect that local knowledge to broader scarcity.

3.2 The price vector as compressed information

Let $\mathbf{p} = (p_1, \dots, p_n)$ be the price vector. Hayek's core insight can be summarized as:

$$\mathbf{p} \approx \text{compressed index of relative scarcity and urgency.}$$

Individuals do not need to know why p_j changed. They only need to know that the relative difficulty or urgency of using good j has changed.

Interpretation: Prices economize on knowledge. They transmit only what is necessary for adjustment, not the full causal story.

3.3 Tin example

Suppose tin becomes scarcer because either:

$$\text{new demand for tin rises} \quad \text{or} \quad \text{a tin supply source disappears.}$$

Most users of tin do not need to know which event occurred. They only need to observe:

$$p_{\text{tin}} \uparrow.$$

They then economize on tin or shift toward substitutes.

Interpretation: The same price signal coordinates many dispersed adjustments: tin users, substitute producers, suppliers, inventories, and downstream products.

3.4 Economy of knowledge

The price system works by transmitting:

$$\text{essential signal} \ll \text{full information.}$$

This is why Hayek compares it to:

$$\text{machinery for registering change} \quad \text{or} \quad \text{a system of telecommunications.}$$

Interpretation: The price mechanism is not valuable because it makes everyone fully informed. It is valuable because it lets everyone act appropriately while remaining mostly uninformed.

3.5 Spontaneous order

The price system was not deliberately designed. Hayek argues that many social institutions work because people follow rules, signals, and practices whose full significance they do not understand.

Interpretation: This is not a claim that existing institutions are perfect. It is a claim that some social orders solve coordination problems that no single planner could consciously solve.

4 Methodological implications

4.1 Why equilibrium can mislead

Equilibrium analysis often asks what allocation follows if all facts are known. Hayek asks a different question:

$$\text{How can people coordinate when each person knows only a fraction of the facts?}$$

Interpretation: The hard problem is not deriving the solution from known facts. The hard problem is communicating and using dispersed facts.

4.2 The ambiguity of “data”

For the observing economist, data may be collected in a model. For the actual economy, those “data” are scattered across many minds.

data given to economist $\neq$ data given to society.

Interpretation: Treating these as the same assumes away the central problem.

4.3 Critique of central planning

Hayek’s critique is informational:

- central planners cannot possess all local and temporary knowledge;
- statistical information loses the particular details needed for action;
- delay in communication matters because economic problems are problems of adaptation;
- decentralized agents can use local knowledge immediately if prices provide broader signals.

Interpretation: The critique is not merely that planners have bad motives. It is that the relevant knowledge is structurally dispersed.

4.4 Critique of monopoly/industry planning

Hayek also criticizes the “half-way house” of delegating planning to organized industries:

industry-level planning $\approx$ monopoly.

Interpretation: Industry planning may reduce decentralization without achieving complete knowledge. It can suppress competitive discovery while still failing to solve the knowledge problem.

5 Identification and empirical design

This is a theoretical essay, not an empirical paper. Its credibility comes from conceptual consistency and explanatory power rather than econometric identification.

5.1 What identifies the argument

- The argument begins from the observable fact that knowledge is dispersed among individuals.
- It distinguishes local/tacit knowledge from scientific/technical knowledge.
- It then asks which institutional mechanism best uses each type of knowledge.
- It identifies the price system as a communication mechanism that transmits relative scarcity.

Interpretation: The logic is comparative institutional analysis: compare central planning, monopoly/industry planning, and competition by their ability to use dispersed knowledge.

5.2 What the essay explains

The essay explains why:

- prices can coordinate behavior without central commands;
- local knowledge is economically important even when it is not scientific;
- market processes can be useful even when participants do not understand the whole system;
- equilibrium models can miss the actual function of prices;
- socialist calculation is difficult because factor values depend on dispersed supply and demand knowledge.

5.3 What the essay does not prove

- It does not estimate the efficiency of markets relative to planning.
- It does not show that all market outcomes are optimal.
- It does not claim prices are always flexible or accurate.
- It does not directly analyze externalities, market power, public goods, or distribution.
- It does not provide a formal dynamic model of information transmission.

Interpretation: The paper's contribution is foundational: it reframes economics as a problem of dispersed knowledge and communication.

6 Extracted original panels

The paper has no empirical tables or graph-style figures. To preserve the original visual material, this section directly extracts key text panels from the paper and places adjacent ideas together to save space.

6.1 Panel 1 + Panel 2: the knowledge problem and decentralized planning

worked out and can be stated best in mathematical form: put at their briefest, they are that the marginal rates of substitution between any two commodities or factors must be the same in all their different uses.

This, however, is emphatically *not* the economic problem which society faces. And the economic calculus which we have developed to solve this logical problem, though an important step toward the solution of the economic problem of society, does not yet provide an answer to it. The reason for this is that the “data” from which the economic calculus starts are never for the whole society “given” to a single mind which could work out the implications, and can never be so given.

The peculiar character of the problem of a rational economic order is determined precisely by the fact that the knowledge of the circumstances of which we must make use never exists in concentrated or integrated form, but solely as the dispersed bits of incomplete and frequently contradictory knowledge which all the separate individuals possess. The economic problem of society is thus not merely a problem

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(a) The “data” are not given to a single mind.

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among many individuals. Planning in the specific sense in which the term is used in contemporary controversy necessarily means central planning—direction of the whole economic system according to one unified plan. Competition, on the other hand, means decentralized planning by many separate persons. The half-way house between the two, about which many people talk but which few like when they see it, is the delegation of planning to organized industries, or, in other words, monopoly.

Which of these systems is likely to be more efficient depends mainly on the question under which of them we can expect that fuller use will be made of the existing knowledge. And this, in turn, depends on whether we are more likely to succeed in putting at the disposal of a single central authority all the knowledge which ought to be used but which is initially dispersed among many different individuals, or in conveying to the individuals such additional knowledge as they need in order to enable them to fit their plans in with those of others.

III

(b) Central planning versus decentralized planning.

Read:

- The first panel states the core thesis: the economic problem is not a logical problem of allocation from complete data.
- The second panel frames planning as a choice between centralized planning and decentralized planning.
- The central criterion is which system can make fuller use of existing knowledge.

Economic message: Hayek shifts the debate from “planning versus no planning” to “which planning system uses dispersed knowledge better.”

6.2 Panel 3 + Panel 4: local knowledge and statistics

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which use can be made only if the decisions depending on it are left to him or are made with his active coöperation. We need to remember only how much we have to learn in any occupation after we have completed our theoretical training, how big a part of our working life we spend learning particular jobs, and how valuable an asset in all walks of life is knowledge of people, of local conditions, and special circumstances. To know of and put to use a machine not fully employed, or somebody's skill which could be better utilized, or to be aware of a surplus stock which can be drawn upon during an interruption of supplies, is socially quite as useful as the knowledge of better alternative techniques. And the shipper who earns his living from using otherwise empty or half-filled journeys of tramp-steamers, or the estate agent whose whole knowledge is almost exclusively one of temporary opportunities, or the *arbitrageur* who gains from local differences of commodity prices, are all performing eminently useful functions based on special knowledge of circumstances of the fleeting moment not known to others.

It is a curious fact that this sort of knowledge should today be generally regarded with a kind of contempt, and that anyone who by such knowledge gains an advantage over somebody better equipped with theoretical or technical knowledge is thought to have acted almost disreputably. To gain an advantage from better knowledge of facilities

(a) Knowledge of local conditions and fleeting opportunities.

and therefore cannot be conveyed to any central authority in statistical form. The statistics which such a central authority would have to use would have to be arrived at precisely by abstracting from minor differences between the things, by lumping together, as resources of one kind, items which differ as regards location, quality, and other particulars, in a way which may be very significant for the specific decision. It follows from this that central planning based on statistical information by its nature cannot take direct account of these circumstances of time and place, and that the central planner will have to find some way or other in which the decisions depending on them can be left to the "man on the spot."

V

If we can agree that the economic problem of society is mainly one of rapid adaptation to changes in the particular circumstances of time and place, it would seem to follow that the ultimate decisions must be left to the people who are familiar with these circumstances, who know directly of the relevant changes and of the resources immediately available to meet them. We cannot expect that this problem will be solved by first communicating all this knowledge to a central board which, after integrating *all* knowledge, issues its orders. We must solve it by some form of decentralization. But this answers only part of our problem. We need decentralization because only thus can we ensure that the knowledge of the particular circumstances of time and place will be promptly used. But the "man on the spot" cannot decide

(b) Statistics abstract from local details; the man on the spot must decide.

Read:

- Local knowledge includes underused machines, particular skills, inventories, transport opportunities, and temporary price differences.
- This knowledge often cannot be summarized in statistics without losing the details that matter.
- The "man on the spot" must use local knowledge promptly, but still needs signals about broader scarcity.

Economic message: Local knowledge is economically productive even when it is not scientific. Prices make local knowledge socially useful.

6.3 Panel 5 + Panel 6: the price system as communication

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Fundamentally, in a system where the knowledge of the relevant facts is dispersed among many people, prices can act to coordinate the separate actions of different people in the same way as subjective values help the individual to coordinate the parts of his plan. It is worth contemplating for a moment a very simple and commonplace instance of the action of the price system to see what precisely it accomplishes. Assume that somewhere in the world a new opportunity for the use of some raw material, say tin, has arisen, or that one of the sources of supply of tin has been eliminated. It does not matter for our purpose—and it is very significant that it does not matter—which of these two causes has made tin more scarce. All that the users of tin need to know is that some of the tin they used to consume is now more profitably employed elsewhere, and that in consequence they must economize tin. There is no need for the great majority of them even to know where the more urgent need has arisen, or in favor of what other needs they ought to husband the supply. If only some of them know directly of the new demand, and switch resources over to it and if the people who are aware of the new gap thus created in turn fill it from still other sources, the effect will rapidly spread throughout the whole economic system and influence not only all the uses of tin, but also those of its substitutes and the substitutes of these substitutes, the supply of all the things made of tin, and their substitutes, and so on; and all this without the great majority of those instrumental in bringing about these substitutions knowing anything at all about the original cause of these changes. The whole acts as one market, not because any of its members survey the whole field, but because their limited individual fields of vision sufficiently overlap so that through many intermediaries the relevant information is communicated to all. The mere fact that there is one price for any commodity—or rather that local prices are connected in a manner determined by the cost of transport, etc.—brings about the solution which (it is just conceptually possible) might have been arrived at by one single mind possessing all the information which is in fact dispersed among all the people involved in the process.

VI

We must look at the price system as such a mechanism for communicating information if we want to understand its real function—a function which, of course, it fulfills less perfectly as prices grow more rigid. (Even when quoted prices have become quite rigid, however, the

(a) The tin example: prices coordinate without explaining causes.

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with which it operates, or how little the individual participants need to know in order to be able to take the right action. In abbreviated form, by a kind of symbol, only the most essential information is passed on, and passed on only to those concerned. It is more than a metaphor to describe the price system as a kind of machinery for registering change, or a system of telecommunications which enables individual producers to watch merely the movement of a few pointers, as an engineer might watch the hands of a few dials, in order to adjust their activities to changes of which they may never know more than is reflected in the price movement.

Of course, these adjustments are probably never “perfect” in the sense in which the economist conceives of them in his equilibrium analysis. But I fear that our theoretical habits of approaching the problem with the assumption of more or less perfect knowledge on the part of almost everyone has made us somewhat blind to the true function of the price mechanism and led us to apply rather misleading standards in judging its efficiency. The marvel is that in a case like that of a scarcity of one raw material, without an order being issued, without more than perhaps a handful of people knowing the cause, tens of thousands of people whose identity could not be ascertained by months of investigation, are made to use the material or its products more sparingly; *i.e.*, they move in the right direction. This is enough of a marvel even if, in a constantly changing world, not all will hit it off so perfectly that their profit rates will always be maintained at the same constant or “normal” level.

I have deliberately used the word “marvel” to shock the reader out of the complacency with which we often take the working of this mechanism for granted. I am convinced that if it were the result of deliberate human design, and if the people guided by the price changes understood that their decisions have significance far beyond their immediate aim, this mechanism would have been acclaimed as one of the greatest triumphs of the human mind. Its misfortune is the double one that it is not the product of human design and that the people guided by it usually do not know why they are made to do what they do. But those who clamor for “conscious direction”—and who cannot believe that anything which has evolved without design (and even without our understanding it) should solve problems which we should not be able to solve consciously—should remember this: The problem is precisely how to extend the span of our utilization of resources beyond the span of the control of any one mind; and, therefore, how to dispense with the need of conscious control and how to provide inducements which will make the individuals do the desirable things without anyone

(b) The price system as telecommunications and as a “marvel.”

Read:

- The tin example shows that users need only know that tin is scarcer, not why it is scarcer.
- Price changes transmit just enough information for substitution and conservation.
- Hayek calls the price system a telecommunications mechanism because it registers change and induces adjustment.

Economic message: The price system economizes on knowledge. It coordinates action precisely by avoiding the need to communicate full information.

6.4 Panel 7 + Panel 8: spontaneous order and the single-mind fallacy

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The problem which we meet here is by no means peculiar to economics but arises in connection with nearly all truly social phenomena, with language and most of our cultural inheritance, and constitutes really the central theoretical problem of all social science. As Alfred Whitehead has said in another connection, "It is a profoundly erroneous truism, repeated by all copy-books and by eminent people when they are making speeches, that we should cultivate the habit of thinking what we are doing. The precise opposite is the case. Civilization advances by extending the number of important operations which we can perform without thinking about them." This is of profound significance in the social field. We make constant use of formulas, symbols and rules whose meaning we do not understand and through the use of which we avail ourselves of the assistance of knowledge which individually we do not possess. We have developed these practices and institutions by building upon habits and institutions which have proved successful in their own sphere and which have in turn become the foundation of the civilization we have built up.

The price system is just one of those formations which man has learned to use (though he is still very far from having learned to make the best use of it) after he had stumbled upon it without understanding it. Through it not only a division of labor but also a coördinated utilization of resources based on an equally divided knowledge has become possible. The people who like to deride any suggestion that this may be so usually distort the argument by insinuating that it asserts that by some miracle just that sort of system has spontaneously grown up which is best suited to modern civilization. It is the other way round: man has been able to develop that division of labor on which our civilization is based because he happened to stumble upon a method which made it possible. Had he not done so he might still have developed some other, altogether different, type of civilization, something like the "state" of the termite ants, or some other altogether unimaginable type. All that we can say is that nobody has yet succeeded in designing an alternative system in which certain features of the existing one can be preserved which are dear even to those who most violently assail it—such as particularly the extent to which the indi-

(a) The price system as a formation not deliberately designed.

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implied in, or follows necessarily from, the valuation of consumers' goods. But this, too, is not correct. Implication is a logical relationship which can be meaningfully asserted only of propositions simultaneously present to one and the same mind. It is evident, however, that the values of the factors of production do not depend solely on the valuation of the consumers' goods but also on the conditions of supply of the various factors of production. Only to a mind to which all these facts were simultaneously known would the answer necessarily follow from the facts given to it. The practical problem, however, arises precisely because these facts are never so given to a single mind, and because, in consequence, it is necessary that in the solution of the problem knowledge should be used that is dispersed among many people.

The problem is thus in no way solved if we can show that all the facts, *if* they were known to a single mind (as we hypothetically assume them to be given to the observing economist), would uniquely determine the solution; instead we must show how a solution is produced by the interactions of people each of whom possesses only partial knowledge. To assume all the knowledge to be given to a single mind in the same manner in which we assume it to be given to us as the explaining economists is to assume the problem away and to disregard everything that is important and significant in the real world.

That an economist of Professor Schumpeter's standing should thus have fallen into a trap which the ambiguity of the term "datum" sets to the unwary can hardly be explained as a simple error. It suggests rather than there is something fundamentally wrong with an approach which habitually disregards an essential part of the phenomena with which we have to deal: the unavoidable imperfection of man's knowledge and the consequent need for a process by which knowledge is constantly communicated and acquired. Any approach, such as that of much of mathematical economics with its simultaneous equations, which in effect starts from the assumption that people's *knowledge* corresponds with the objective *facts* of the situation, systematically leaves out what is our main task to explain. I am far from denying that in our system equilibrium analysis has a useful function to perform. But when it comes to the point where it misleads some of our

(b) The practical problem arises because facts are not given to one mind.

Read:

- The price system is one of many social formations that people use without fully understanding.
- Hayek rejects the idea that showing a solution would follow if all facts were known solves the actual economic problem.
- The actual problem is how interaction among people with partial knowledge generates a solution.

Economic message: The final methodological lesson is that assuming complete knowledge removes exactly what economics must explain.

7 Short summary

- Hayek argues that the economic problem of society is the use of dispersed knowledge.
- The standard optimization problem assumes all relevant data are already given, but society never has such data in one mind.
- Local knowledge of time and place is essential and often cannot be captured by statistics.
- All economic activity is planning; the real question is whether planning is centralized or decentralized.
- Central planning struggles because it cannot collect, update, and use local knowledge quickly enough.
- The price system transmits compressed information about relative scarcity.

- People do not need to know why a price changes; they only need to know how the relative terms of action have changed.
- Prices coordinate countless adjustments without central commands.
- This coordination is imperfect but valuable because it economizes on knowledge.
- Hayek's methodological critique is that equilibrium analysis can obscure the real function of markets by assuming away ignorance.
- The essay is a foundation for later work on information, coordination, spontaneous order, and market process economics.

8 Conclusion

8.1 Core conclusion

Hayek's core conclusion is that economic organization must be judged by how well it uses dispersed knowledge. Since relevant knowledge is held by many individuals in fragmentary and local form, the price system has a unique role as a communication mechanism. It allows individuals to coordinate their plans without requiring any single mind to know the whole economic situation.

8.2 Economic intuition

The intuition is that every person sees only a tiny part of the economy. A central planner would need to gather and update millions of local facts, many of which are temporary, tacit, or impossible to express statistically. Prices compress these facts into signals of relative scarcity. When a price changes, people adjust their behavior even if they do not know the underlying cause. This makes decentralized coordination possible.

8.3 Takeaway

The enduring lesson is that prices are not merely incentives or equilibrium numbers. They are carriers of knowledge. The market's value lies in allowing society to use knowledge that no one possesses as a whole.